

Convergencia casi segura implica en probabilidad

Proposición 1. Sea $(X_n)_{n \in \mathbb{N}}$ tal que $X_n \xrightarrow{c.s.} X \implies X_n \xrightarrow{\mathbb{P}} X$.

Demostración: Sea $\varepsilon > 0$, tenemos que $\exists A \in \Sigma : \mathbb{P}(A) = 0 : \forall \omega \in A^c : X_n(\omega) \rightarrow X(\omega)$.

Para $\omega \in A^c$, $\limsup_{n \rightarrow \infty} |X_n(\omega) - X(\omega)| = 0$

$$\implies \mathbb{P} \left(\left\{ \omega \in \Omega : \limsup_{n \rightarrow \infty} |X_n(\omega) - X(\omega)| > \varepsilon \right\} \right) = 0.$$

Por otro lado, por el Lema de Fatou para probabilidades, tenemos que

$$\begin{aligned} \mathbb{P} \left(\left\{ \omega \in \Omega : \limsup_{n \rightarrow \infty} |X_n(\omega) - X(\omega)| > \varepsilon \right\} \right) &\geq \limsup_{n \rightarrow \infty} \mathbb{P}(\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\}) \\ &\geq \lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) \geq 0. \end{aligned}$$

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